

BETTLES, AK, USA

WMO: 701740

Lat: 66.917N

Lon: 151.515W

Elev: 634

StdP: 14.36

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26533

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-49.3	-44.2	-35.6	0.7	-28.9	-33.3	0.9	-27.6	18.1	10.6	15.9	11.3	0.6	30	0.721

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	78.9	59.5	75.5	58.0	72.3	56.8	61.9	74.1	60.2	71.3	58.7	68.8	6.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.5	72.1	63.5	55.7	67.7	62.6	54.0	63.6	61.6	27.9	74.1	26.8	71.2	25.7	68.4	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	12.6	11.2	-52.5	84.0	4.9	2.9	-56.0	86.2	-58.9	87.9	-61.6	89.5	-65.1	91.7
			-36.5	64.9	6.3	2.5	-41.1	66.8	-44.8	68.2	-48.3	69.7	-52.9	71.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.4	-10.3	-2.2	3.8	24.8	45.2	58.3	59.7	52.8	41.8	21.8	0.7	-5.9	(6)
	DBStd	28.31	20.19	15.90	13.27	11.98	9.63	6.09	5.72	6.25	7.88	11.57	14.89	15.86	(7)
	HDD50	10119	1869	1460	1433	756	205	8	5	40	257	875	1478	1733	(8)
	HDD65	14866	2334	1880	1898	1206	613	211	181	382	695	1340	1928	2198	(9)
	CDD50	757	0	0	0	0	57	258	304	126	12	0	0	0	(10)
	CDD65	30	0	0	0	0	1	11	16	2	0	0	0	0	(11)
	CDH74	442	0	0	0	0	19	167	218	38	0	0	0	0	(12)
	CDH80	56	0	0	0	0	1	20	32	3	0	0	0	0	(13)
	WSAvg	4.9	4.1	4.7	5.3	6.0	5.8	5.1	4.9	4.5	4.7	4.8	4.4	4.2	(14)
	PrecAvg	14.3	0.8	0.7	0.7	0.6	0.7	1.4	2.1	2.4	1.9	1.1	1.0	0.9	(15)
(16) Precipitation	PrecMax	24.9	3.4	3.2	3.6	3.1	3.0	3.6	5.4	5.9	4.8	3.8	3.9	3.1	(16)
	PrecMin	6.9	0.0	0.0	0.0	0.0	0.1	0.1	0.3	0.6	0.1	0.1	0.1	0.1	(17)
	PrecStd	4.0	0.8	0.7	0.7	0.6	0.6	0.7	1.2	1.2	1.3	0.7	0.9	0.6	(18)
	PrecStd	4.0	0.8	0.7	0.7	0.6	0.6	0.7	1.2	1.2	1.3	0.7	0.9	0.6	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.8	33.4	38.7	54.8	76.6	82.5	83.8	78.8	65.5	49.2	33.6	30.8	(19)
		MCWB	29.2	31.6	32.4	42.5	55.3	60.0	61.4	62.5	50.9	43.5	30.4	29.3	(20)
	2%	DB	27.4	29.1	33.3	49.4	70.8	78.5	79.4	73.0	61.7	42.4	29.1	25.3	(21)
		MCWB	26.1	27.0	29.6	39.5	52.5	57.9	60.5	59.0	49.8	37.8	27.5	24.0	(22)
	5%	DB	22.5	24.9	28.8	45.6	66.1	75.3	76.0	68.3	58.1	39.0	25.4	21.0	(23)
		MCWB	21.4	23.3	25.2	37.1	50.0	56.5	59.6	56.4	48.3	35.6	24.2	20.0	(24)
	10%	DB	17.5	19.9	24.1	42.1	61.9	72.1	72.5	64.3	54.1	35.6	20.5	15.7	(25)
		MCWB	16.3	18.4	21.0	34.8	48.3	55.6	58.4	54.5	46.9	32.9	19.4	14.7	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.0	31.8	33.6	43.1	56.0	62.5	64.2	64.2	53.3	42.3	30.9	29.6	(27)
		MCDB	31.9	32.9	37.1	53.5	71.9	76.2	76.4	74.9	62.4	45.1	32.8	31.1	(28)
	2%	WB	26.7	27.4	30.0	40.0	53.7	60.3	62.4	60.4	51.1	38.8	27.9	24.3	(29)
		MCDB	28.0	29.1	33.0	48.0	67.6	73.1	74.8	70.0	58.3	41.6	29.1	25.4	(30)
	5%	WB	22.7	23.6	25.5	37.6	51.4	58.6	61.0	58.2	49.6	35.7	24.4	20.4	(31)
		MCDB	23.7	25.2	28.7	44.7	64.1	71.2	72.6	65.6	56.2	38.3	25.8	21.3	(32)
	10%	WB	18.1	19.0	21.1	35.2	49.2	56.9	59.6	56.0	47.8	33.2	19.5	15.3	(33)
		MCDB	19.1	20.3	24.1	41.4	60.7	69.2	70.0	62.4	53.5	35.3	20.5	16.2	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	14.6	17.6	23.8	20.8	19.8	20.7	18.9	18.0	15.6	12.6	12.8	13.8	(35)
		MCDBR	14.8	16.9	22.9	20.1	25.8	26.3	26.1	24.3	22.9	14.3	12.9	13.8	(36)
		MCWBR	13.8	15.2	18.9	12.5	11.9	11.0	11.0	12.4	13.1	10.0	11.6	12.9	(37)
	5% WB	MCDBR	14.7	16.8	21.7	18.5	23.1	23.2	22.0	19.8	18.9	12.2	12.4	13.2	(38)
		MCWBR	13.7	15.2	18.1	11.8	11.1	10.5	10.1	10.8	11.3	8.9	11.3	12.4	(39)
(40) Clear-Sky Solar Irradiance	taub		0.168	0.232	0.256	0.310	0.351	0.361	0.367	0.357	0.313	0.265	0.170	N/A	(40)
	taud		1.919	2.154	2.250	2.177	2.296	2.374	2.383	2.426	2.492	2.228	1.934	N/A	(41)
	Ebn at Noon		71	199	259	267	267	268	261	252	239	178	62	0	(42)
	Edh at Noon		4	15	24	34	34	33	31	27	20	14	3	0	(43)
(44) All-Sky Solar Radiation	RadAvg		24	208	780	1472	1737	1801	1510	1114	664	246	50	0	(44)
	RadStd		2	17	49	83	154	141	113	114	89	37	6	0	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+1.51	N/A	N/A	N/A	N/A	N/A	N/A	-493	N/A	N/A
(47)	Regional (3 neighbors)	+1.73	N/A	N/A	N/A	N/A	N/A	N/A	-603	N/A	N/A

Nomenclature: See separate page